

Packages

Packages and **Imports** are Java's system for organizing code file folders and accessing tools.

Think of a large computer. You do not throw every single picture, document, and song onto your desktop. You organize them into folders like /Images or /Music.

- **Packages** are the **folders** used to group related Java files together.
- **Rules for Packages:**
 - **The Declaration:** If a file belongs in a package folder, the very first line of code *must* state that package name.
 - **Naming Convention:** Package names are always completely lowercase. Large companies use their reversed website domain to make them unique (e.g., com.apple.shapes)
- **Example of a Package File:**

Imagine you create a folder named geometry. Inside it, you make a file named Circle.java. The code looks like this:

```
package geometry; // 1. Declares that this file lives in the geometry folder
public class Circle {
    public double calculateArea(double radius) {
        return 3.14 * radius * radius;
    }
}
```

The Built-In Java Packages

Java comes with thousands of pre-made folders filled with tools ready for you to use.

Here are the three most common folders beginner programmers pull tools from: 

Package Folder Name	What It Contains	Example Tools
<code>java.util</code>	Utilities and helpful developer tools.	<code>Scanner</code> (reading keyboards), <code>Random</code> (generating random numbers).
<code>java.time</code>	Tools for tracking dates and clocks.	<code>LocalDate</code> (calendar dates), <code>LocalTime</code> (clock hours).
<code>java.io</code>	Input and Output tools.	<code>File</code> (reading text files saved on your computer hard drive).

imports

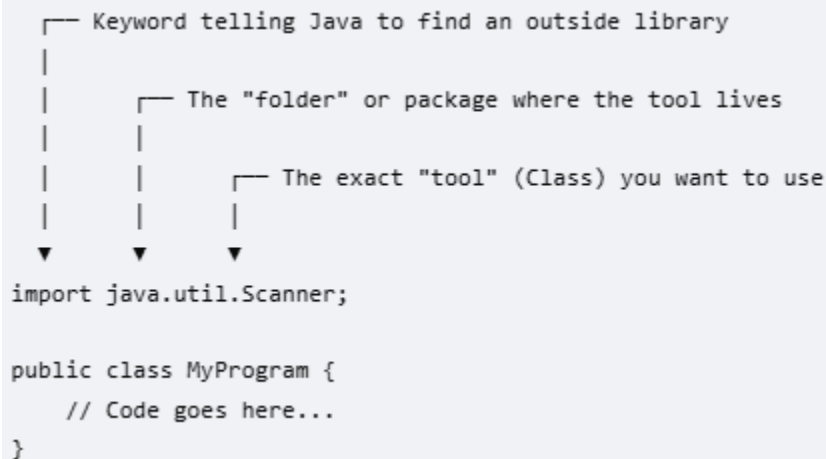
If you want to use that Circle tool inside a completely different project folder, your main program will not be able to find it automatically. You must **import** it to bridge the gap.

- import keyword is used to bring pre-written code libraries into your program from massive Java digital libraries.
- By default Java only loads a tiny handful of essential libraries (like basic math and text strings) so your program stays lightweight and fast.
- Place import statements at the very top of your code file, before your class even begins.
- Importing Your Custom Package:

```
import geometry.Circle; // 1. Pulls the Circle tool out of the geometry folder
public class MainProgram {
    public static void main(String[] args) {
```

```
        Circle myShape = new Circle(); // 2. Now you can use it freely!
```

```
    }
}
```



The diagram illustrates the components of the import statement `import java.util.Scanner;` with arrows pointing from descriptive text to the corresponding parts of the code:

- `import`: Keyword telling Java to find an outside library
- `java`: The "folder" or package where the tool lives
- `util`: The "folder" or package where the tool lives
- `Scanner`: The exact "tool" (Class) you want to use

```
import java.util.Scanner;

public class MyProgram {
    // Code goes here...
}
```